Exam Two — November 19

The second exam covers tanks, buildings, 1st-order linear equations, 1st-order separable equations, 1st-order constant-coefficient linear systems, and variation of parameters.

- This includes y_c, y_p, and initial conditions.
- The exam will be like HW (with simpler boxes) stuck together.
- Four questions, on both sides of two sheets of paper, totaling 50 points.
- You can write anything you want on a notecard up to 4×6 inches.

The exam is intended to be very predictable.

Finding y_c for $y' = Ay + f(t)$

- Real eigenpairs $\boldsymbol{A}\boldsymbol{v}_i = \lambda_i \boldsymbol{v}_i$ with n independent eigenvectors

$$\boldsymbol{y}_c(t) = c_1 e^{\lambda_1 t}\boldsymbol{v}_1 + \cdots + c_n e^{\lambda_n t}\boldsymbol{v}_n$$

- Complex eigenvalue $\alpha + \beta i$ and eigenvector $\boldsymbol{a} + \boldsymbol{b}i$

$$\begin{aligned}\boldsymbol{y}_c(t) = \quad & c_1 e^{\alpha t}\big(\boldsymbol{a}\cos(\beta t) - \boldsymbol{b}\sin(\beta t)\big) \\ & + c_2 e^{\alpha t}\big(\boldsymbol{a}\sin(\beta t) + \boldsymbol{b}\cos(\beta t)\big)\end{aligned}$$

- Double eigenvalue λ_1 with only one eigenvector $\boldsymbol{v}_1$

$$(\boldsymbol{A} - \lambda_1 \boldsymbol{I})\boldsymbol{w} = \boldsymbol{v}_1 \qquad \text{solutions} \qquad c_1 e^{\lambda_1 t}\boldsymbol{v}_1 + c_2 e^{\lambda_1 t}(t\boldsymbol{v}_1 + \boldsymbol{w})$$

Finding y_p for $y' = Ay + f(t)$

- Undetermined coefficients. Guess the form of $\boldsymbol{y}_p$ from $\boldsymbol{f}$ and substitute into the ODE then match coefficients.
  - Polynomials. $\boldsymbol{f}(t) = \begin{pmatrix} t^2+1 \\ 2t \end{pmatrix} \longrightarrow \boldsymbol{y}_p = \boldsymbol{a}_2 t^2 + \boldsymbol{a}_1 t + \boldsymbol{a}_0$
  - Exponentials. $\boldsymbol{f}(t) = e^{2t}\begin{pmatrix} 1 \\ 3 \end{pmatrix} \longrightarrow \boldsymbol{y}_p = \boldsymbol{a}e^{2t}$
  - Sinusoids. $\boldsymbol{f}(t) = \begin{pmatrix} \sin(2t) \\ \cos(2t) \end{pmatrix} \longrightarrow \boldsymbol{y}_p = \boldsymbol{a}\cos(2t) + \boldsymbol{b}\sin(2t)$
- Variation of parameters. Form the fundamental matrix $\Phi(t) = [\boldsymbol{y}_1(t) \ \cdots \ \boldsymbol{y}_n(t)]$
  - Let $\boldsymbol{y}_p(t) = \Phi(t)\boldsymbol{u}(t)$ and solve

$$\boldsymbol{u}'(t) = \Phi(t)^{-1}\boldsymbol{f}(t) \qquad \boldsymbol{y}_p(t) = \Phi(t)\left(\int^t \Phi(s)^{-1}\boldsymbol{f}(s)\,ds + \boldsymbol{C}\right)$$